Pokémon Win Percentage Statistical Analysis

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Abstract

This paper presents a statistical analysis of Pokémon win percentages, focusing on the impact of various Pokémon characteristics and abilities. The dataset, sourced from Kaggle, provides comprehensive attributes of each Pokémon and extensive battle information. The primary research question is: "Is there a significant difference in the win percentage of a Pokémon based on its Type, Number of Types, Generation, and Legendary Status?"

Introduction

This study conducts a detailed statistical analysis to understand the factors influencing the win percentage of a Pokémon. The dataset, titled "Pokémon Data Analysis Tutorial" and sourced from Kaggle, is composed of three files: pokemon.csv, combats.csv, and tests.csv. The win percentage for each Pokémon was computed using the battle information from the combats.csv file. The study aims to determine whether there is a significant difference in win percentages based on a Pokémon's Type 1 (there are 18 unique types), Number of Types (whether it has 1 or 2 types), Generation (the season of the show the Pokemon came out), and Legendary Status. The significance of this study lies in its potential to provide insights into the factors that contribute most to a Pokémon's success in battles.

Data Description

The dataset comprises three files: pokemon.csv, combats.csv, and tests.csv. We will focus on using pokemon.csv and combats.csv.

pokemon.csv

This file contains detailed attributes of each Pokémon, including:

- Record Number
- Pokemon Name
- 1st Attack Type
- 2nd Attack Type
- Hitpoints
- Attack Force
- Defense Points
- Special Attack Force
- Special Defense Points
- Speed of Pokemon
- Generation
- Legendary Status (1 = legendary, 0 = ordinary)

combats.csv

This file provides information on several battles between two Pokémon, including the winner of each battle.

Data Exploration

Our first step involves reading and familiarizing ourselves with the data. To enhance readability, we will adjust column names as necessary. We will then explore the dimensions of each data frame to understand the structure of our dataset. Given our interest in comparing Pokémon performance based on their types, we will specifically examine the unique values in the 'Type 1' column. Lastly, we will ensure the integrity of our data by checking for missing values.

```
pokemon = read.csv("~/Downloads/pokemon.csv")
names(pokemon)[names(pokemon) == 'X.'] <- 'Number'

combats = read.csv("~/Downloads/combats.csv")

head(pokemon)

##   Number         Name Type.1 Type.2 HP Attack Defense Sp..Atk Sp..Def Speed
## 1      1     Bulbasaur  Grass Poison 45     49      49      65      65    45
## 2      2       Ivysaur  Grass Poison 60     62      63      80      80    60
## 3      3      Venusaur  Grass Poison 80     82      83     100     100    80
## 4      4 Mega Venusaur  Grass Poison 80    100     123     122     120    80
## 5      5    Charmander   Fire        39     52      43      60      50    65
## 6      6    Charmeleon   Fire        58     64      58      80      65    80
##   Generation Legendary
## 1          1     False
## 2          1     False
## 3          1     False
## 4          1     False
## 5          1     False
## 6          1     False

dim(pokemon) #there are 800 Pokemon

## [1] 800  12

head(combats)

##   First_pokemon Second_pokemon Winner
## 1           266           298    298
## 2           702           701    701
## 3           191           668    668
## 4           237           683    683
## 5           151           231    151
## 6           657           752    657

dim(combats) #there are approximately 50,000 battles between Pokemon

## [1] 50000     3

unique(pokemon$Type.1) ; length(unique(pokemon$Type.1)) #number of types

##  [1] "Grass"    "Fire"     "Water"    "Bug"      "Normal"   "Poison"
##  [7] "Electric" "Ground"   "Fairy"    "Fighting" "Psychic"  "Rock"
## [13] "Ghost"    "Ice"      "Dragon"   "Dark"     "Steel"    "Flying"
```

```
## [1] 18

colSums(is.na(pokemon)) #there are no missing values

##     Number       Name     Type.1     Type.2         HP     Attack    Defense
##          0          0          0          0          0          0          0
##    Sp..Atk    Sp..Def      Speed Generation  Legendary
##          0          0          0          0          0
```

Data Preparation

Our data preparation phase begins with the calculation of the win percentage for each Pokémon, which will be represented in a new column.

```
first_count = table(combats$First_pokemon)
second_count = table(combats$Second_pokemon)

total_battles = first_count + second_count #total battles per Pokemon

total_wins = table(combats$Winner) # total wins per Pokemon

length(as.vector(total_wins))

## [1] 783

length(as.vector(total_battles))

## [1] 784

total_battles = as.data.frame(total_battles)
total_wins = as.data.frame(total_wins)
```

We noticed a discrepancy in the dimensions between total wins and total battles, suggesting there is a Pokémon that did not win any battles.

```
setdiff(total_battles$Var1, total_wins$Var1)

## [1] "231"

print(pokemon[231,])

##     Number    Name Type.1 Type.2 HP Attack Defense Sp..Atk Sp..Def Speed
## 231    231 Shuckle    Bug   Rock 20     10     230      10     230     5
##     Generation Legendary
## 231          2     False
```

To rectify this, we identified the Pokémon - ‘Shuckle’, and included it in the total wins to align the dimensions.

```
shuckle <- data.frame(Var1 = 231, Freq = 0)
total_wins <- rbind(shuckle, total_wins)

colnames(total_battles) = c("Number", "Battles")
colnames(total_wins) = c("Number", "Wins")

#Calculating win percentage
df = merge(total_battles,total_wins, by = "Number")
df$`Win Percentage` = df$Wins/df$Battles

pokemon <- merge(pokemon, df, by = "Number")
```

Analysis

Comparison of Pokemon with only Type 1 (18 unique types).

Our analysis investigates whether there is a statistically significant difference in the mean win percentage among Pokémon that have only one type. With 18 unique types in our dataset, this comparison will provide insights into the influence of Pokémon type on battle outcomes.

```
two_types = pokemon[Reduce(`&`,
                    lapply(pokemon, function(x) !(is.na(x)|x==""))),]

one_type = pokemon[!Reduce(`&`,
                    lapply(pokemon, function(x) !(is.na(x)|x==""))),]

two_types = two_types[, c("Number", "Name", "Type.1", "Type.2",
                    "Win Percentage")]

one_type = one_type[, c("Number", "Name", "Type.1",  "Win Percentage")]
```

Assumption Checking

Normality and Homogeneity of Variances

```
library(car)

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode

library(biotools)

## Loading required package: MASS

##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select

## ---
## biotools version 4.2

# Normality
for(type in unique(one_type$Type.1)) {
  subset <- one_type[one_type$Type.1 == type,]

  # Check if the sample size is within the valid range for Shapiro-Wilk test
  if(nrow(subset) >= 3 & nrow(subset) <= 5000) {
    shapiro_test <- shapiro.test(subset$`Win Percentage`)
    print(paste("Type:", type, "p-value:", shapiro_test$p.value))
  } else {
    print(paste("Type:", type, "invalid sample size for Shapiro-Wilk test"))
  }
}

## [1] "Type: Fire p-value: 0.275612674301564"
## [1] "Type: Water p-value: 0.156520098048428"
## [1] "Type: Bug p-value: 0.0014946515934276"
## [1] "Type: Normal p-value: 0.00567513138863092"
## [1] "Type: Poison p-value: 0.194818568299475"
## [1] "Type: Electric p-value: 0.0233971653437067"
## [1] "Type: Ground p-value: 0.0516519344076551"
## [1] "Type: Fairy p-value: 0.28018696247991"
## [1] "Type: Fighting p-value: 0.163390354551413"
## [1] "Type: Psychic p-value: 0.00315816040432053"
## [1] "Type: Grass p-value: 0.0107381975762248"
## [1] "Type: Dragon p-value: 0.267342855734345"
## [1] "Type: Rock p-value: 0.403267606920838"
## [1] "Type: Dark p-value: 0.221670179849004"
## [1] "Type: Ghost p-value: 0.179734059611949"
## [1] "Type: Steel p-value: 0.224219662694562"
## [1] "Type: Ice p-value: 0.512643138706428"
## [1] "Type: Flying invalid sample size for Shapiro-Wilk test"

# Homogeneity of variances
levene_test <- leveneTest(`Win Percentage` ~ Type.1, data = one_type)

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

print(levene_test)

## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group  17  2.3854 0.001667 **
##       360
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Our results indicate that the win percentages for the Pokémon types Bug, Normal, Electric, Psychic, and Grass have p-values less than 0.05, implying a deviation from a normal distribution.

The Flying type, with only two Pokémon, has an insufficient sample size for the test, which requires at least three per group.

```
sum(one_type$Type.1 == "Flying") # number of Pokemon that fly

## [1] 2
```

Furthermore, our p-value of 0.001667 on Levene's Test for Homogeneity, being less than 0.05, leads us to reject the null hypothesis and infer that variances are not homogeneous across different Pokémon types.

Testing if there is a significant statistical difference in win percentage among all pokeon with one type

Initially, we considered using ANOVA for our analysis. However, due to our data not meeting the assumptions of normality and homogeneity of variances, we opted for a non-parametric alternative, the Kruskal-Wallis test.

ANOVA

```
#ANOVA function for pokemon with one type
attach(one_type)

one_type$Type.1 <- as.factor(one_type$Type.1)
one.lm = lm(`Win Percentage`~ Type.1, data = one_type)
anova(one.lm)

## Analysis of Variance Table
##
## Response: Win Percentage
##           Df  Sum Sq  Mean Sq F value   Pr(>F)
## Type.1     17  3.4668 0.203928  3.3564 9.73e-06 ***
## Residuals 360 21.8732 0.060759
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

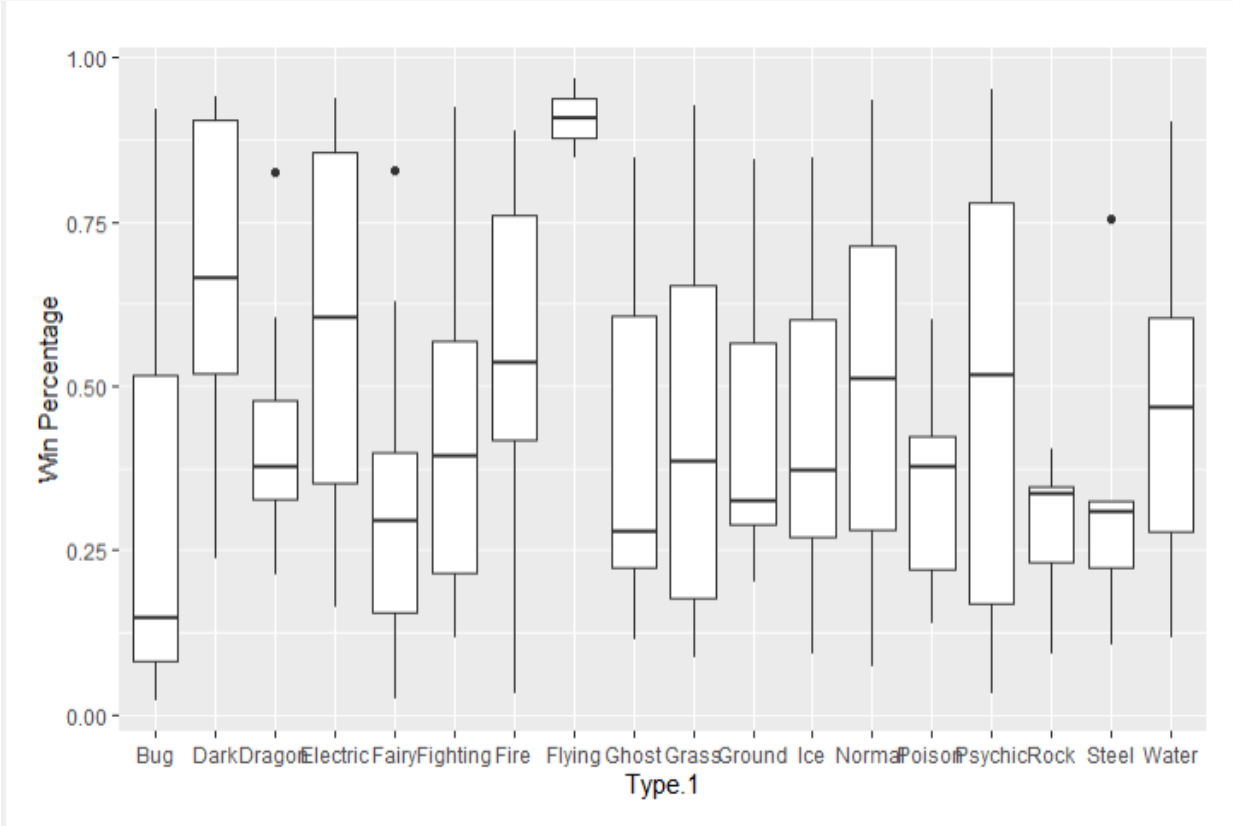
Our analysis suggests that at least one of the mean win percentages among the Pokémon types is unequal. Hence, we will use the Pairwise T-Test to reveal statistically significant differences between the 18 unique Pokemon types.

```
#Pairs t-test
pairwise.t.test(`Win Percentage`, Type.1, "bon")

##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  Win Percentage and Type.1
##
##          Bug    Dark   Dragon Electric Fairy  Fighting Fire   Flying Ghost
## Dark     0.0126 -      -      -       -      -       -      -      -
## Dragon   1.0000 1.0000 -      -       -      -       -      -      -
## Electric 0.0042 1.0000 1.0000 -       -      -       -      -      -
## Fairy    1.0000 0.0608 1.0000 0.0399  -      -       -      -      -
## Fighting 1.0000 0.7475 1.0000 0.8580  1.0000 -       -      -      -
## Fire     0.0534 1.0000 1.0000 1.0000  0.3529 1.0000  -      -      -
## Flying   0.1296 1.0000 1.0000 1.0000  0.2515 0.9959  1.0000 -      -
## Ghost    1.0000 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 -
## Grass    1.0000 0.7416 1.0000 0.6389  1.0000 1.0000  1.0000 1.0000 1.0000
## Ground   1.0000 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 1.0000
## Ice      1.0000 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 1.0000
## Normal   0.1420 1.0000 1.0000 1.0000  0.9970 1.0000  1.0000 1.0000 1.0000
## Poison   1.0000 0.1952 1.0000 0.1793  1.0000 1.0000  1.0000 0.4504 1.0000
## Psychic  0.6583 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 1.0000
## Rock     1.0000 0.0917 1.0000 0.1042  1.0000 1.0000  0.5948 0.2121 1.0000
## Steel    1.0000 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 1.0000
## Water    1.0000 1.0000 1.0000 1.0000  1.0000 1.0000  1.0000 1.0000 1.0000
##          Grass  Ground Ice    Normal Poison Psychic Rock   Steel
## Dark     -      -      -      -      -      -      -      -
## Dragon   -      -      -      -      -      -      -      -
## Electric -      -      -      -      -      -      -      -
## Fairy    -      -      -      -      -      -      -      -
## Fighting -      -      -      -      -      -      -      -
## Fire     -      -      -      -      -      -      -      -
## Flying   -      -      -      -      -      -      -      -
## Ghost    -      -      -      -      -      -      -      -
## Grass    -      -      -      -      -      -      -      -
## Ground   1.0000 -      -      -      -      -      -      -
## Ice      1.0000 1.0000 -      -      -      -      -      -
## Normal   1.0000 1.0000 1.0000 -      -      -      -      -
## Poison   1.0000 1.0000 1.0000 1.0000 -      -      -      -
## Psychic  1.0000 1.0000 1.0000 1.0000 1.0000 -      -      -
## Rock     1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 -      -
## Steel    1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 -
## Water    1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
##
## P value adjustment method: bonferroni
```

According to the Pairwise T-Test, there is a statistically significant difference between the Fairy type versus the Electric type, and the Bug type versus Dark, Electric, and Fire types, indicating that the Bug and Fairy types significantly differ from the other types.

Furthermore, the Bug type is the most significantly different from all the other types This suggests that Bug type Pokémon tend to lose more battles.

```
library(tidyverse)

qplot(data = one_type, x = Type.1, y = `Win Percentage`, geom = "boxplot")
```



The box plot further illustrates the low win percentage of Bug type Pokémon. It's noteworthy that the mean win percentage of the Flying type Pokémon is exceptionally high due to the small sample size (only two observations), making it an outlier.

Kruskal-Wallis

```
#Kruskal-Wallis Test

kruskal_test <- kruskal.test(`Win Percentage` ~ Type.1, data = one_type)
print(kruskal_test)

##
##  Kruskal-Wallis rank sum test
```

```
##
## data:  Win Percentage by Type.1
## Kruskal-Wallis chi-squared = 50.219, df = 17, p-value = 3.906e-05
```

The p-value we obtained is less than 0.05, which suggests that we can reject the null hypothesis and conclude that there is a statistically significant difference in the Win Percentage across the different Pokemon types.

However, the Kruskal-Wallis test does not tell us which specific groups (in this case, Pokemon types) were significantly different from each other. To find out, we can perform the Dunn test.

```
#Dunns Test

library(dunn.test)
dunn_result <- dunn.test(one_type$`Win Percentage`, one_type$Type.1,
                         method = "bonferroni")

##   Kruskal-Wallis rank sum test
##
## data: x and group
## Kruskal-Wallis chi-squared = 50.2188, df = 17, p-value = 0
##
##
##                              Comparison of x by group
##                                   (Bonferroni)
## Col Mean-|
## Row Mean |       Bug      Dark     Dragon   Electric      Fairy   Fighting
## ---------+-----------------------------------------------------------------
##     Dark |  -3.876505
##          |    0.0081*
##          |
##   Dragon |  -1.662088   2.063804
##          |     1.0000     1.0000
##          |
## Electric |  -4.111039   0.734923  -1.760389
##          |    0.0030*     1.0000     1.0000
##          |
##    Fairy |  -0.465042   3.380667   1.205191   3.440958
##          |     1.0000     0.0553     1.0000     0.0443
##          |
## Fighting |  -1.508501   2.665293   0.368303   2.568806  -0.981112
##          |     1.0000     0.5884     1.0000     0.7807     1.0000
##          |
##     Fire |  -3.651668   1.145817  -1.347870   0.556326  -2.994150  -2.083121
##          |    0.0199*     1.0000     1.0000     1.0000     0.2105     1.0000
##          |
##   Flying |  -3.036265  -0.935782  -2.116017  -1.360362  -2.796336  -2.375776
##          |     0.1832     1.0000     1.0000     1.0000     0.3954     1.0000
```

```
##          |
##    Ghost |  -1.164329   2.416909   0.409980   2.184898  -0.733075   0.101349
##          |    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##    Grass |  -1.863070   2.701936   0.230459   2.699304  -1.263914  -0.201325
##          |    1.0000     0.5274     1.0000     0.5316     1.0000     1.0000
##          |
##   Ground |  -1.756831   2.061077  -0.046093   1.759484  -1.284928  -0.430604
##          |    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##      Ice |  -1.625144   2.249348   0.108342   1.996768  -1.145390  -0.264360
##          |    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##   Normal |  -3.306230   1.856229  -0.812839   1.561152  -2.577641  -1.541006
##          |    0.0723     1.0000     1.0000     1.0000     0.7610     1.0000
##          |
##   Poison |  -0.935252   2.972655   0.785574   2.923709  -0.456170   0.498854
##          |    1.0000     0.2259     1.0000     0.2646     1.0000     1.0000
##          |
##  Psychic |  -2.754850   2.085097  -0.469295   1.863352  -2.095851  -1.068477
##          |    0.4492     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##     Rock |  -0.131976   3.243937   1.309875   3.165570   0.261682   1.110108
##          |    1.0000     0.0902     1.0000     0.1184     1.0000     1.0000
##          |
##    Steel |  -0.494716   2.361059   0.725794   2.097396  -0.168369   0.501221
##          |    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##    Water |  -2.735767   2.300944  -0.342733   2.212145  -2.037558  -0.952819
##          |    0.4761     1.0000     1.0000     1.0000     1.0000     1.0000
## Col Mean-|
## Row Mean |       Fire     Flying      Ghost      Grass     Ground        Ice
## ---------+--------------------------------------------------------------
##   Flying |  -1.567054
##          |    1.0000
##          |
##    Ghost |   1.788199   2.331185
##          |    1.0000     1.0000
##          |
##    Grass |   2.149890   2.340417  -0.270169
##          |    1.0000     1.0000     1.0000
##          |
##   Ground |   1.334331   2.104521  -0.463301  -0.294483
##          |    1.0000     1.0000     1.0000     1.0000
##          |
##      Ice |   1.561353   2.199947  -0.320355  -0.110449   0.158936
##          |    1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##   Normal |   0.926760   1.891043  -1.304423  -1.568028  -0.782225  -1.016156
##          |    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
```

```
##   Poison |   2.473570   2.575061   0.325064   0.734318   0.854846   0.705813
##          |     1.0000     0.7667     1.0000     1.0000     1.0000     1.0000
##          |
##  Psychic |   1.280626   2.020627  -0.956113  -0.998090  -0.427132  -0.638212
##          |     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##     Rock |   2.788192   2.833873   0.891490   1.341282   1.378781   1.255359
##          |     0.4055     0.3518     1.0000     1.0000     1.0000     1.0000
##          |
##    Steel |   1.794203   2.412042   0.387661   0.644457   0.771580   0.659552
##          |     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
##          |
##    Water |   1.589233   2.104134  -0.851702  -0.868197  -0.294792  -0.511639
##          |     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
## Col Mean-|
## Row Mean |     Normal     Poison    Psychic       Rock      Steel
## ---------+-------------------------------------------------------
##   Poison |   2.001601
##          |     1.0000
##          |
##  Psychic |   0.510921  -1.549594
##          |     1.0000     1.0000
##          |
##     Rock |   2.391180   0.656737   2.021581
##          |     1.0000     1.0000     1.0000
##          |
##    Steel |   1.413581   0.154191   1.160629  -0.353693
##          |     1.0000     1.0000     1.0000     1.0000
##          |
##    Water |   0.829628  -1.463556   0.228281  -1.956075  -1.081297
##          |     1.0000     1.0000     1.0000     1.0000     1.0000
##
## alpha = 0.05
## Reject Ho if p <= alpha/2
```

The results from the Dunns test indicate that there are statistically significant differences in win percentages between Bug type and Dark, Electric, and Fire type.

```
# Calculate median win percentage for Bug type
bug_median <- median(one_type$`Win Percentage`[one_type$Type.1 == "Bug"])

# Calculate median win percentage for Dark type
dark_median <- median(one_type$`Win Percentage`[one_type$Type.1 == "Dark"])

# Calculate median win percentage for Electric type
electric_median <- median(one_type$`Win Percentage`[one_type$Type.1 == "Electric"])

# Calculate median win percentage for Fire type
fire_median <- median(one_type$`Win Percentage`[one_type$Type.1 == "Fire"])
```

```
# Print the results
print(paste("Median win percentage for Bug type:", bug_median))

## [1] "Median win percentage for Bug type: 0.146341463414634"

print(paste("Median win percentage for Dark type:", dark_median))

## [1] "Median win percentage for Dark type: 0.664510112490529"

print(paste("Median win percentage for Electric type:", electric_median))

## [1] "Median win percentage for Electric type: 0.604477611940298"

print(paste("Median win percentage for Fire type:", fire_median))

## [1] "Median win percentage for Fire type: 0.536337955829481"
```

Based on the median win percentages we have calculated, it appears that Bug types have a lower median win percentage (0.146) compared to Dark (0.665), Electric (0.604), and Fire (0.536) types. This suggests that, on average, Bug type Pokemon tend to win less often when battling against these types.

Findings: Comparison of Pokemon with only Type 1 (18 unique types)

Initially, we planned to use an ANOVA test to determine if there was a statistically significant difference in the win percentages across the different Pokemon types. However, upon checking the assumptions for the ANOVA test, we found they were not met. Specifically, the assumption of normality was violated for several Pokemon types, and the assumption of homogeneity of variances was also not met.

As a result, we turned to the non-parametric Kruskal-Wallis test, which does not require these assumptions. The results from the Kruskal-Wallis test were consistent with what we would have expected from the ANOVA test, had its assumptions been met. We found that there is a statistically significant difference in the win percentages across the different Pokemon types.

Further analysis using Dunn's test revealed that the Bug type Pokemon showed statistically significant differences in win percentages when compared to other types. This suggests that the performance of Bug type Pokemon in battles is significantly different from that of several other types.

It's important to note that while the assumptions for the ANOVA test were not met, leading us to use the Kruskal-Wallis test instead, both tests led us to the same conclusion about the overall differences in win percentages across Pokemon types. This demonstrates the robustness of our findings.

Comparison of Pokemon number of Types (1 Type versus 2 Types)

Our analysis investigates whether there is a statistically significant difference in the mean win percentage among Pokémon with only one type versus Pokemon with two different types. This comparison will provide insights into the influence of Pokémon's number of types on battle outcomes.

Assumption Checking

Normality

```
library(ICSNP)

## Loading required package: mvtnorm

## Loading required package: ICS

# Convert vectors to one-column matrices
one_type_matrix <- matrix(one_type$`Win Percentage`, ncol = 1)
two_types_matrix <- matrix(two_types$`Win Percentage`, ncol = 1)
```

```
shapiro.test(one_type_matrix) #Test for Normal data for Pokemon with one type
```

```
        Shapiro-Wilk normality test

data:  one_type_matrix
W = 0.95478, p-value = 2.258e-09
```

From the Shapiro-Wilk normality test for Pokemon with one type, the p-value is essentially zero which means we reject the null hypothesis. Thus, the data for the win percentage data for Pokemon that have just one type are not normally distributed.

```
shapiro.test(two_types_matrix) #Test for Normal data for Pokemon with two types
```

```
        Shapiro-Wilk normality test

data:  two_types_matrix
W = 0.96659, p-value = 5.331e-08
```

From the Shapiro-Wilk normality test for Pokemon with two types, the p-value is essentially zero which means we reject the null hypothesis. Thus, the data for the win percentage data for Pokemon with two types are not normally distributed.

Hotellings T2 and Mann-Whitney U Test

We originally used Hotelling's T2 for our analysis, but due to the data not meeting the assumptions of normality, we decided to explore a non-parametric alternative, such as the Mann-Whitney U test.

```
# Perform HotellingsT2 test
HotellingsT2(one_type_matrix, two_types_matrix)

##
##  Hotelling's two-sample T2-test
##
## data:  one_type_matrix and two_types_matrix
## T.2 = 14.005, df1 = 1, df2 = 782, p-value = 0.0001956
## Alternative hypothesis: true location difference is not equal to c(0)

# Perform the Mann-Whitney U test
mw_test = wilcox.test(one_type$`Win Percentage`, two_types$`Win Percentage`)
mw_test

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  one_type$`Win Percentage` and two_types$`Win Percentage`
## W = 64853, p-value = 0.0001771
## Alternative hypothesis: true location shift is not equal to 0
```

The results from both, the Hotelling's T2 test and the Mann-Whitney U test, indicate a statistically significant difference in the Win Percentage between Pokemon with one type and those with two types.

The test statistic for Hotelling's T2 is 14.005, and the p-value is 0.0001956. Since the p-value is less than 0.05, we can reject the null hypothesis and conclude there is a statistically significant difference in the Win Percentage between Pokemon with one type and those with two types.

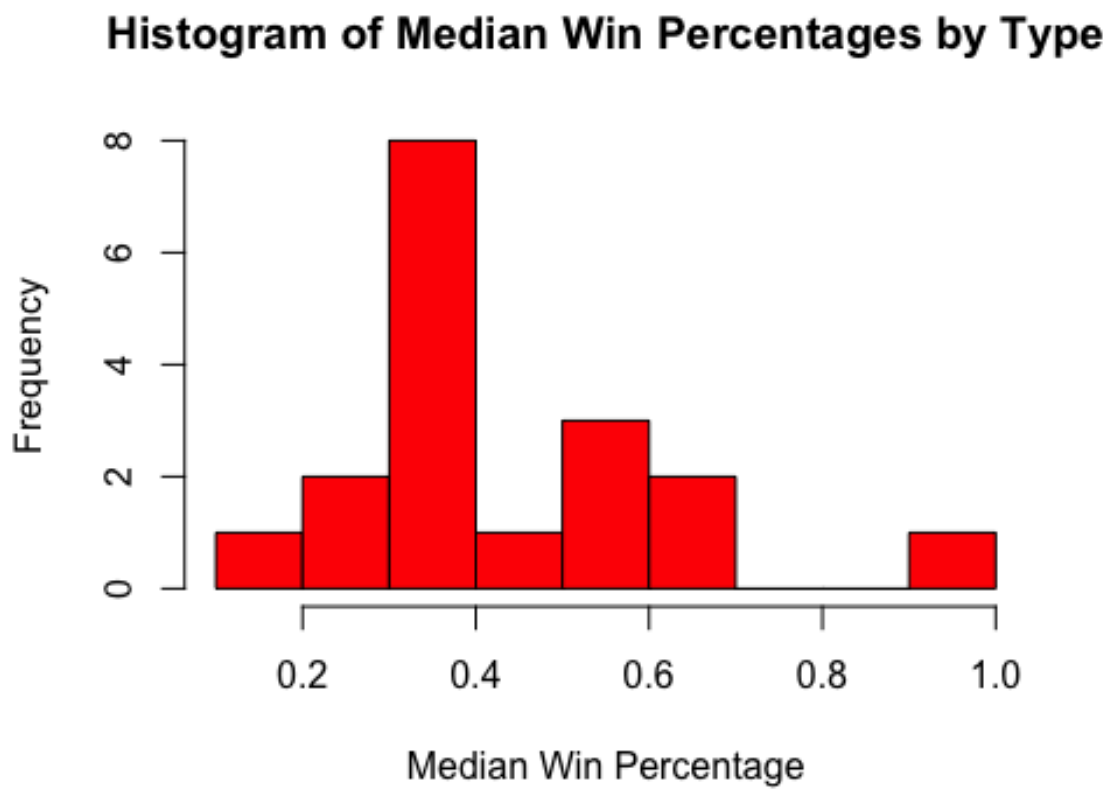
The test statistic for Mann-Whitney is 64853, and the p-value is 0.0001771. Again, since the p-value is less than 0.05, we can reject the null hypothesis and conclude that there is a

statistically significant difference in the Win Percentage between Pokemon with one type and those with two types.

In conclusion, both tests suggest that the number of types a Pokemon has (one type vs. two types) significantly affects its win percentage. However, these tests do not tell us which group (one type or two types) tends to have a higher win percentage. For that, we could look at the median win percentages of the two groups.

Median win percentage for Pokemon of just one type

```
#The medians of Pokemon with just one type (by type)
unique_groups <- unique(one_type$Type.1)

for (group_name in unique_groups) {
  group_median <- median(one_type$`Win Percentage`[one_type$Type.1 == group_name], na.rm =
TRUE)
  cat("Type:", group_name, "- Median Value:", group_median, "\n")
}

## Type: Fire - Median Value: 0.536338
## Type: Water - Median Value: 0.4662162
## Type: Bug - Median Value: 0.1463415
## Type: Normal - Median Value: 0.511811
## Type: Poison - Median Value: 0.3759398
## Type: Electric - Median Value: 0.6044776
## Type: Ground - Median Value: 0.3241979
## Type: Fairy - Median Value: 0.2941176
## Type: Fighting - Median Value: 0.392
## Type: Psychic - Median Value: 0.5152311
## Type: Grass - Median Value: 0.384
## Type: Dragon - Median Value: 0.3777778
## Type: Rock - Median Value: 0.3358779
## Type: Dark - Median Value: 0.6645101
## Type: Ghost - Median Value: 0.2794118
## Type: Steel - Median Value: 0.3097345
## Type: Ice - Median Value: 0.372093
## Type: Flying - Median Value: 0.9074234

medians <- numeric(length(unique_groups))

# Calculating the medians of each single type
for (i in seq_along(unique_groups)) {
  group_name <- unique_groups[i]
  group_median <- median(one_type$`Win Percentage`
                         [one_type$Type.1 == group_name], na.rm = TRUE)
  medians[i] <- group_median
}

# Plot histogram
hist(medians, main = "Histogram of Median Win Percentages by Type",
     xlab = "Median Win Percentage", col = "red")
```



After calculating the median win percentage for Pokemon of one type and viewing the histogram, it's difficult to identify a central tendency. The distribution is skewed to the right, showing greater variability. This means that there are possible outliers. From the output of the medians, the flying type has a median win percentage of 90.74%. Earlier, we found only two Pokemon in the data set are just the flying type. This explains why the distribution is extremely skewed.

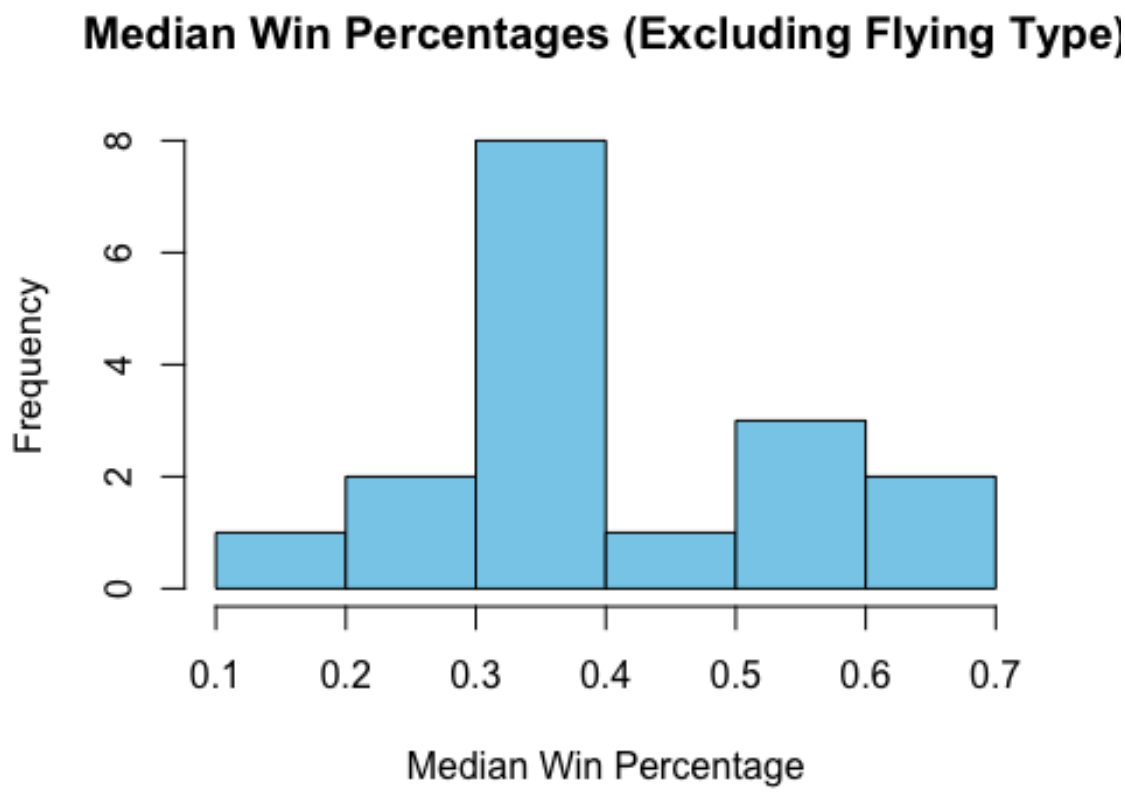
Furthermore, 9 of the 18 types fall within the 30-40% median win percentage range. Ice, Steel, Rock, Dragon, Grass, Fighting, Fairy, Ground, and Poison types fall within this range. Excluding the Flying, Dark, Psychic, Electric, Normal, and Fire types all have a median win percentage above 50%. This goes to show which Pokemon types win more often than others.

```
one_type_counts <- table(one_type$Type.1)
one_type_counts

##
##     Bug    Dark   Dragon Electric    Fairy Fighting     Fire   Flying
##      17      10       11       27       15       19       28        2
##   Ghost   Grass   Ground      Ice   Normal   Poison  Psychic     Rock
##      10      31       12       13       59       15       38        9
```

```
##    Steel    Water
##        5       57
```

For further investigation, the Pokemon types that make up most of the population are Normal, Psychic, Fire, Water, Electric, and Fighting. Besides the Dark type, these types all have a median win percentage above 50%. Having a higher sample size for each Pokemon type could explain the robustness of the median win percentage, but it may not mean that the win percentage would be higher. Pokemon types vary in their strengths and weaknesses, which influences their likelihood of winning battles. Some types may naturally have higher win percentages due to their effectiveness against other types, regardless of sample size.

```
# removing Flying-type Pokémon
one_type_no_flying <- one_type[one_type$Type.1 != "Flying", ]

overall_one_median <- median(one_type_no_flying$`Win Percentage`,
                             na.rm = TRUE)
print(overall_one_median)

## [1] 0.4338684

unique_types <- unique(one_type_no_flying$Type.1)
type_medians <- numeric(length(unique_types))

for (i in seq_along(unique_types)) {
  type <- unique_types[i]
  type_medians[i] <- median(one_type_no_flying$`Win Percentage`
                            [one_type_no_flying$Type.1 == type], na.rm = TRUE)
}

# plotting the distribution of median win percentages
hist(type_medians,
     main = "Median Win Percentages (Excluding Flying Type)",
     xlab = "Median Win Percentage",
     col = "skyblue",
     border = "black")
```

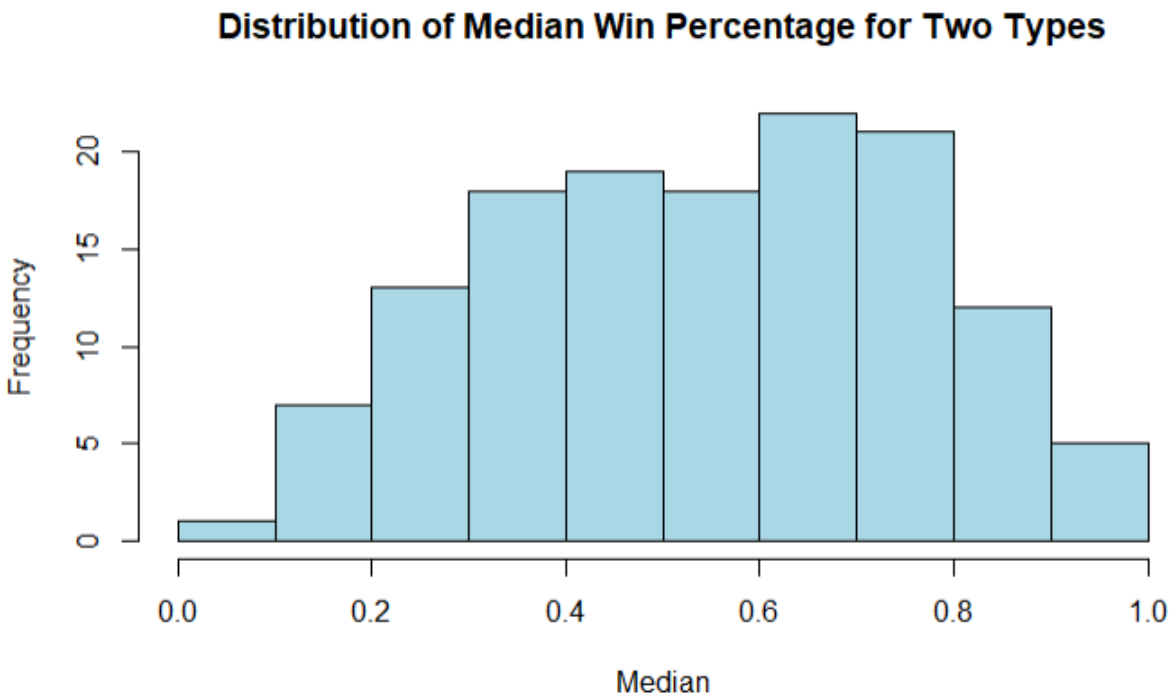


The distribution appears to be less skewed when excluding the two Flying type observations and recreating the histogram. The distribution is still not normal, but for comparison, we decided to find the central tendency. The overall median of the median win percentages for each type is 43.38%.

Median win percentage for Pokemon of two types

```
result <- aggregate(two_types$`Win Percentage`~ two_types$Type.1 +
                    two_types$Type.2, data = two_types, FUN = median,
                   na.rm = TRUE)
head(result) #first 10

##   two_types$Type.1 two_types$Type.2 two_types$`Win Percentage`
## 1          Poison              Bug                 0.4468085
## 2            Rock              Bug                 0.4351786
## 3        Fighting             Dark                 0.3739837
## 4           Ghost             Dark                 0.2457627
## 5           Grass             Dark                 0.4647887
## 6          Ground             Dark                 0.5752212

dim(result)

## [1] 136   3
```

```
hist(result$`two_types$\`Win Percentage\``, breaks = 10, col = "lightblue",
    main = "Distribution of Median Win Percentage for Two Types",
    xlab = "Median", ylab = "Frequency")
```



136 total unique combinations for Pokemon have two types. This means that the number of observations per combination may vary. However, there seems to be a normal distribution of the median win percentage for Pokemon of two types.

Since the distribution of the medians is normal, it's appropriate to calculate the central tendency of the medians.

```
median_win_percentages <- result$`two_types$\`Win Percentage\``

overall_median <- median(median_win_percentages, na.rm = TRUE)

# Print the result
print(overall_median)

## [1] 0.5536283
```

The overall median win percentage of Pokemon with two types is 55.36%. This shows the average performance of Pokemon across various types. Thus, since the median is over 50%, it suggests that on average, Pokemon with type combinations are relatively balanced and perform

well. This is much higher than the overall median win percentage for Pokemon with just one type, which was 43.38%.

Findings: Comparison of Pokemon number of Types (1 Type vs 2 Types)
Our analysis reveals a significant impact of the number of types (one type vs. two types) a Pokémon has on its win percentage. Both Hotelling's T2 test and the Mann-Whitney U test indicate a statistically significant difference in the win percentages between Pokémon with one type and those with two types.

Pokémon with just one type have a median win percentage of 43.38%, with the distribution of win percentages being skewed to the right. This skewness is primarily due to the Flying type, which has only two Pokémon and a high median win percentage of 90.74%. Excluding the Flying type, the median win percentages for most types fall within the 30-40% range, with a few types (Dark, Psychic, Electric, Normal, and Fire) having a median win percentage above 50%.

On the other hand, Pokémon with two types have a higher overall median win percentage of 55.36%, suggesting that on average, these Pokémon are relatively balanced and perform well in battles. The distribution of win percentages for Pokémon with two types appears to be normal, indicating less variability compared to Pokémon with just one type.

In conclusion, the number of types a Pokémon has significantly affected its win percentage, with Pokémon having two types generally performing better than those with just one type. This finding is consistent with the results from both Hotelling's T2 test and the Mann-Whitney U test, demonstrating the robustness of our analysis. However, it’s important to note that the strengths and weaknesses of different Pokémon types, as well as their effectiveness against other types, can also influence their likelihood of winning in battles. Further research could explore these aspects in more detail.

Comparison of Pokemon based on Generation
Our analysis investigates whether there is a statistically significant difference in the mean win percentage among Pokémon belonging to different generations. The generation of a Pokemon represents in which season the Pokemon came out. There are six generations. A Pokemon can belong to only one. This comparison will provide insights into the influence of Pokémon's Generation on battle outcomes.

Assumption Checking
Normality and Homogeneity of Variances

```
pokemon$Generation <- as.factor(pokemon$Generation)

# Normality
for(generation in unique(pokemon$Generation)) {
  subset <- pokemon[pokemon$Generation == generation,]

  # Check if the sample size is within the valid range for the Shapiro-Wilk test
  if(nrow(subset) >= 3 & nrow(subset) <= 5000) {
```

```
  shapiro_test <- shapiro.test(subset$`Win Percentage`)
  print(paste("Generation:", generation, "p-value:", shapiro_test$p.value))
 } else {
  print(paste("Generation:", generation, "has an invalid sample size for Shapiro-Wilk test"))
 }
}

## [1] "Generation: 1 p-value: 9.37896151253172e-05"
## [1] "Generation: 2 p-value: 0.00720078949834128"
## [1] "Generation: 3 p-value: 6.71338294629233e-05"
## [1] "Generation: 4 p-value: 0.000107478920469069"
## [1] "Generation: 5 p-value: 7.49594914292612e-06"
## [1] "Generation: 6 p-value: 0.0566371358714428"

# Homogeneity of variances
levene_test <- leveneTest(`Win Percentage` ~ Generation, data = pokemon)
print(levene_test)

## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  5  0.5312 0.7527
##       778
```

From our Shapiro-Wilk test results, it appears the win percentages for all Pokemon generations except Generation 6 have p-values less than 0.05, suggesting that they do not follow a normal distribution. For Generation 6, the p-value is slightly greater than 0.05. Thus, we reject the null hypothesis and conclude that the generations do not follow a normal distribution.

From our Levene test results, the p-value is 0.7527, which is greater than 0.05. We fail to reject the null hypothesis and conclude that the variances are equal across the different Pokemon generations.

Testing if there is a significant statistical difference in win percentage

As we have found from our previous tests, the win percentages for all Pokemon generations except Generation 6 do not follow a normal distribution, which violates one of the key assumptions of ANOVA. However, the variances across the different Pokemon generations are equal, which meets the homogeneity of variances assumption of ANOVA.

Given these findings, the Kruskal-Wallis test, which does not require the assumption of normality, would be a more appropriate choice in this case.

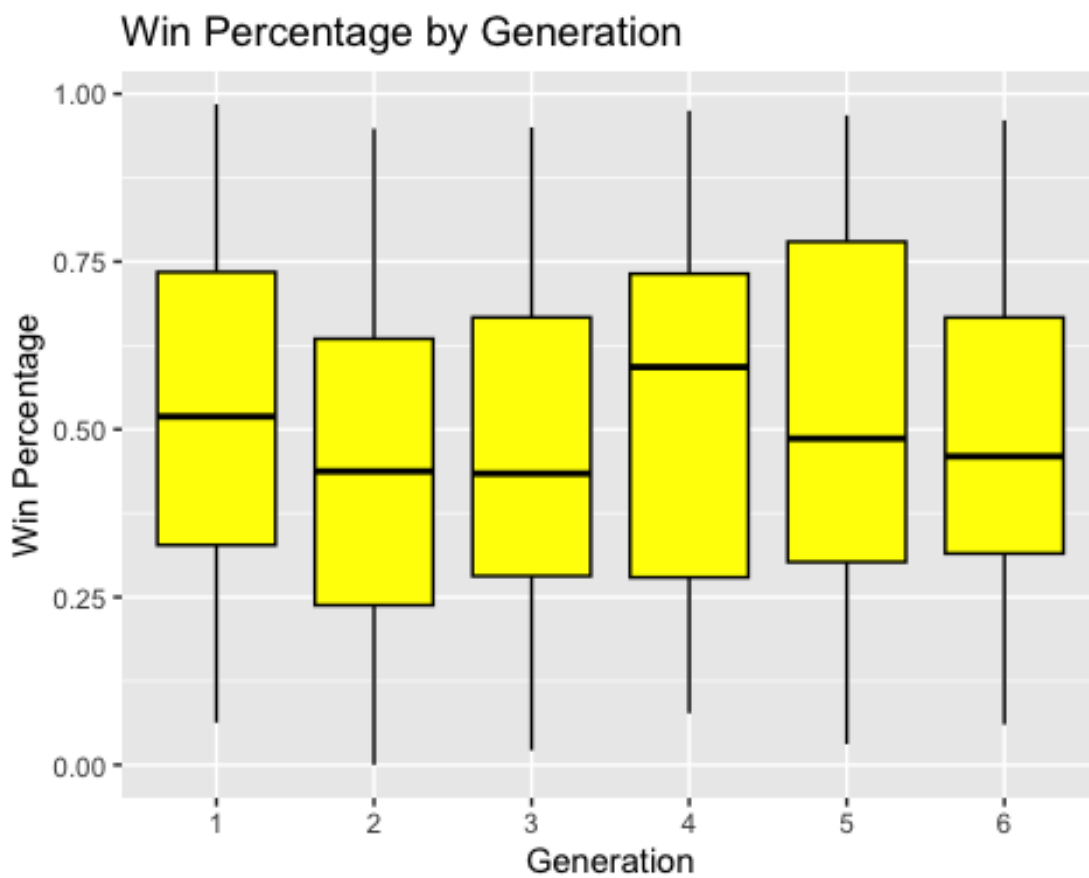
Kruskal-Wallis

```
#Kruskal-Wallis Test

kruskal_test <- kruskal.test(`Win Percentage` ~ Generation, data = pokemon)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Win Percentage by Generation
## Kruskal-Wallis chi-squared = 9.6688, df = 5, p-value = 0.08518
```

The p-value is 0.08518, which is greater than the commonly used significance level of 0.05. This means that we fail to reject the null hypothesis. Our results do not provide strong evidence to conclude that there are statistically significant differences in the Win Percentage across the different Pokemon generations.

```
ggplot(pokemon, aes(x = Generation, y = `Win Percentage`)) +
  geom_boxplot(fill = "yellow", color = "black") +
  labs(title = "Win Percentage by Generation",
       x = "Generation",
       y = "Win Percentage")
```




```
generation_medians <- aggregate(`Win Percentage` ~ Generation, data = pokemon,
                                FUN = median)
print(generation_medians) # median win percentage for each distribution
```

```
##   Generation Win Percentage
## 1          1      0.5190840
## 2          2      0.4374024
## 3          3      0.4340785
## 4          4      0.5928258
## 5          5      0.4862964
## 6          6      0.4596774
```

The box plot shows the median win percentage for each generation. The win percentages for each generation are 0.52, 0.43, 0.43, 0.59, 0.48, and 0.46 respective to their generations. Although the fourth generation has the highest win percentage, there isn't a large enough difference in these values to suggest a significant difference.

Lastly, these results suggest that the generation a Pokemon is from doesn't necessarily impact their likelihood of winning or losing a battle.

Findings: Comparison of Pokemon based on Generation

Our analysis of the impact of Pokémon generations on win percentages reveals that the generation a Pokémon belongs to does not significantly influence its likelihood of winning or losing battles. This conclusion is supported by the Kruskal-Wallis test, which yielded a p-value of 0.08518, greater than the commonly used significance level of 0.05. This led us to fail to reject the null hypothesis, suggesting no strong evidence of statistically significant differences in the Win Percentage across different Pokémon generations.

The median win percentages for each generation, ranging from 0.43 to 0.59, further corroborate this finding. Despite Generation 4 having the highest median win percentage, the differences between the generations are not large enough to suggest a significant impact on battle outcomes.

In conclusion, our findings suggest that while individual Pokémon characteristics such as type and number of types can significantly affect battle performance, the generation a Pokémon belongs to does not appear to have a substantial impact on its win percentage. This insight could be valuable for Pokémon trainers in strategizing their battles, as it indicates that focusing on individual Pokémon characteristics could be more beneficial than considering their generation. However, further research could explore other potential influences on Pokémon battle outcomes.

Pokemon comparisons on Legendary status

Our analysis investigates whether there is a statistically significant difference in the median win percentage among Pokémon with either Legendary or Non-Legendary status. This comparison will provide insights into the influence of Pokémon's Legendary status on battle outcomes.

Assumption Checking

Normality and Homogeneity of Variances

```
#Check assumptions
```

```
pokemon$Legendary <- as.factor(pokemon$Legendary)

# Normality
legendary_pokemons <- pokemon[pokemon$Legendary == "True",]
non_legendary_pokemons <- pokemon[pokemon$Legendary == "False",]

shapiro_test_leg <- shapiro.test(legendary_pokemons$`Win Percentage`)
shapiro_test_non_leg <- shapiro.test(non_legendary_pokemons$`Win Percentage`)

print(paste("Legendary Pokemons p-value:", shapiro_test_leg$p.value))

## [1] "Legendary Pokemons p-value: 1.55275886843077e-06"

print(paste("Non-Legendary Pokemons p-value:", shapiro_test_non_leg$p.value))

## [1] "Non-Legendary Pokemons p-value: 1.30737467489478e-11"

# Homogeneity of variances
levene_test <- leveneTest(`Win Percentage` ~ Legendary, data = pokemon)
print(levene_test)

## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value    Pr(>F)
## group   1  30.442 4.679e-08 ***
##       782
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From our Shapiro-Wilk test results, it appears the win percentages for both Legendary and Non-Legendary Pokemon have p-values less than 0.05, suggesting that they do not follow a normal distribution.

After testing the homogeneity of the variances, we found that the p-value is significantly less than 0.05. We reject the null hypothesis. This suggests the variances are not equal between Legendary and Non-Legendary Pokemon.

Given these findings, the assumptions for an ANOVA test are not met. Therefore, a non-parametric test like the Mann-Whitney U test would be more appropriate for comparing the Win Percentage between Legendary and Non-Legendary Pokemon.

Mann-Whitney

```
# Perform the Mann-Whitney U test
wilcox.test(legendary_pokemons$`Win Percentage`, non_legendary_pokemons$`Win Percentage`)

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  legendary_pokemons$`Win Percentage` and non_legendary_pokemons$`Win Percentage`
```

```
## W = 38514, p-value < 2.2e-16
## Alternative hypothesis: true location shift is not equal to 0
```

The p-value is less than 2.2e-16. The p-value is essentially zero and much smaller than the significance level of 0.05. We reject the null hypothesis and conclude there is very strong evidence of a statistically significant difference in the Win Percentage between Legendary and Non-Legendary Pokemon.

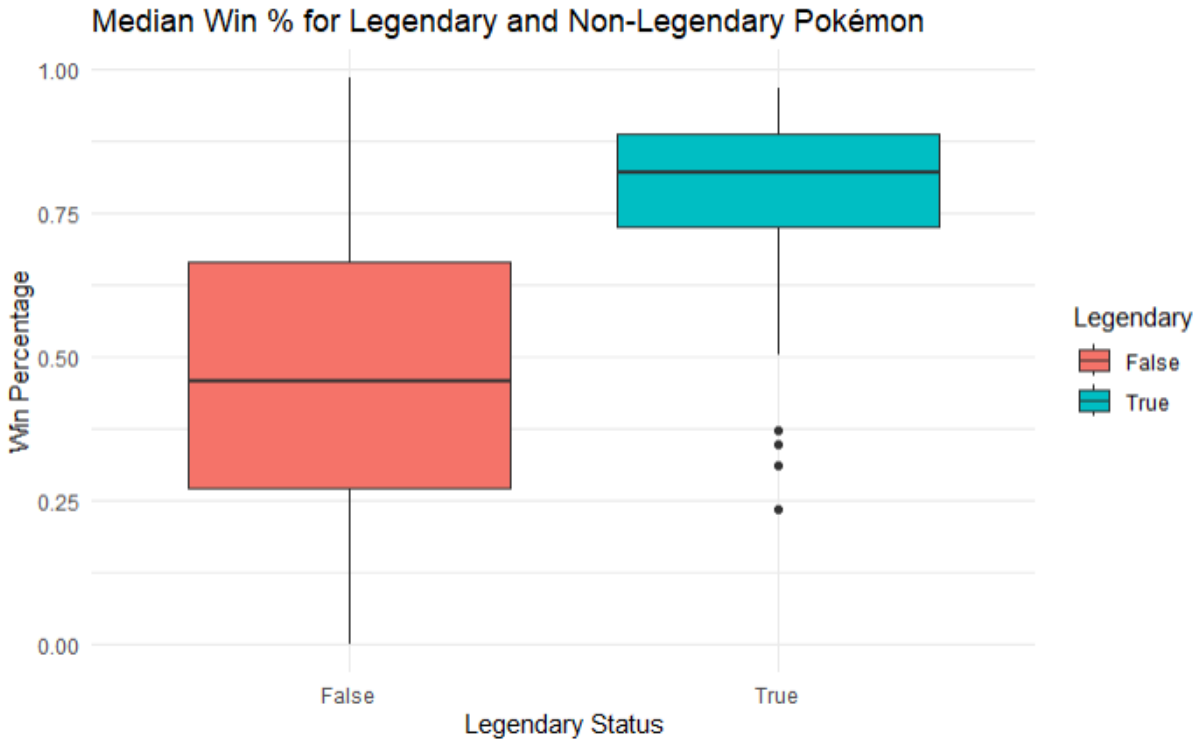
Median win percentage for Legendary and Non-Legendary Pokemon

If we look at the Legendary and Non-Legendary Pokemon win percentages:

```
print(paste("Legendary Pokemons Win Percentage median:",
        median(legendary_pokemons$`Win Percentage`)))

## [1] "Legendary Pokemons Win Percentage median: 0.821002132196162"

print(paste("Non-Legendary Pokemons Win Percentage median:",
        median(non_legendary_pokemons$`Win Percentage`)))

## [1] "Non-Legendary Pokemons Win Percentage median: 0.456389452332657"
```

We can observe that Legendary Pokemon have a median win percentage of approximately 0.821, while Non-Legendary Pokemon have a median win percentage of about 0.456. This suggests that Legendary Pokemon win battles almost twice as often as Non-Legendary Pokemon. This could be due to the typically superior stats and abilities of Legendary Pokemon.

```
ggplot(pokemon, aes(x = Legendary, y = `Win Percentage`, fill = Legendary)) +
  geom_boxplot() +
  labs(title = "Median Win % for Legendary and Non-Legendary Pokémon",
      x = "Legendary Status",
      y = "Win Percentage",
      fill = "Legendary") +
  theme_minimal()
```



Findings: Comparison of Pokemon based on Legendary status

Our analysis reveals a significant impact of a Pokémon's Legendary status on its win percentage. Both the Shapiro-Wilk test and the Mann-Whitney U test indicate a statistically significant difference in the win percentages between Legendary and Non-Legendary Pokémon.

Legendary Pokémon, with a median win percentage of approximately 0.821, tend to win battles almost twice as often as Non-Legendary Pokémon, which have a median win percentage of about 0.456. This significant difference could be attributed to the typically superior stats and abilities of Legendary Pokémon.

In conclusion, our findings suggest that a Pokémon's Legendary status significantly influences its battle performance, with Legendary Pokémon generally outperforming Non-Legendary ones. This insight could be valuable for Pokémon trainers in strategizing their battles, as it indicates that focusing on a Pokémon's Legendary status could be beneficial. However, further research could explore other potential influences on Pokémon battle outcomes.

Conclusion

Our comprehensive statistical analysis of Pokémon win percentages has yielded several key insights. We investigated the influence of various Pokémon characteristics on battle outcomes, including the number of types, generation, and legendary status.

Our findings suggest that the number of types a Pokémon has significantly affects its win percentage. Specifically, Pokémon with two types generally perform better in battles than those with just one type. This could be due to the increased versatility and range of abilities that come with having two types.

When focusing on Pokémon with only one type, we found a statistically significant difference in the win percentages across the different Pokémon types. Notably, Bug type Pokémon showed statistically significant differences in win percentages compared to other types. This suggests the performance in battles of Bug type Pokemon is significantly different from that of several other types.

In contrast, the generation a Pokémon belongs to does not appear to significantly influence its win percentage. Despite the variations in characteristics and abilities across generations, these differences do not translate into significant disparities in battle performance. This suggests that trainers could focus more on individual Pokémon characteristics rather than their generation when strategizing for battles.

Interestingly, a Pokémon's legendary status has a substantial impact on its win percentage. Legendary Pokémon, known for their superior stats and abilities, tend to win battles almost twice as often as Non-Legendary Pokémon. This insight could be particularly valuable for trainers in understanding the potential advantages of Legendary Pokémon in battles.

In conclusion, our study provides a deeper understanding of the factors that contribute to a Pokémon's success in battles. While our findings offer valuable insights, they also highlight the complexity of Pokémon battles, where the outcome can be influenced by a multitude of factors. Further research could explore other potential influences on Pokémon battle outcomes, providing even more strategic insights for Pokémon trainers. Our study underscores the importance of strategic thinking in Pokémon battles, where understanding the strengths and weaknesses of different Pokémon types, generations, and statuses can make the difference between victory and defeat.